



TED UNIVERSITY

Syllabus for EE 222 Digital Logic Design

Spring 2015-2016

Section 1	
Instructor:	Asst. Prof. Tolga İNAN
Office:	321
E-mail:	tolga.inan@tedu.edu.tr
Time Schedule:	Wednesday (14.00 – 15.50) G108 Thursday (14.00 – 15.50) A311L
Office Hour:	Wednesday (11.00 - 11.50)

Section 2	
Instructor:	Dr. Emre ERDİL
E-mail:	eerdil@gmail.com
Time Schedule:	Monday (14.00 – 15.50) G105 Friday (09.00 – 10.50) A311L

TA:	Res. Asst. Selman Fatih Avcı
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Course Code & Number	EE222	Course Title	Digital Logic Design
Type of Course	X Compulsory O Elective	Semester	O Fall X Spring O Summer
Level of Course	BSc	Year of Study	Sophomore
Course Credit Hours / ECTS	(2+0+2) 3 / 6 ECTS	Pre-requisite / Co-requisite	Pre-requisite: None
Mode of Delivery	X Face-to-face O Distance learning	Language of Instruction	X English O Turkish
Course Coordinator	Asst.Prof. Tolga İNAN		
Computer Usage	Xilinx ISE WebPACK Design Software		
Textbook	Digital Design: With an Introduction to the Verilog HDL (M. M. Mano, M. D. Ciletti), 5th ed., Prentice Hall		
Supplementary Reading			
Module and Instructor Evaluation Date	TBD		

Course Catalog Description	Number systems and Boolean Algebra. Binary logic and gates. Combinatorial and sequential circuit design. Registers and counters. Programmable logic.
Course Objectives	This course aims to introduce the fundamentals of number system, Boolean Algebra and digital logic design.
Course Learning Outcomes (LO)	Upon successful completion of this course, a student will be able to 1. Identify numbers systems, signed arithmetic 2. Use Boolean Algebra, recognize Operator precedence 3. Use standard forms and simplify of Boolean functions 4. Construct Karnaugh Maps and develop 2-level implementations 5. Design combinational circuits 6. Operate middle scale integration circuits (Decoders, MUX, etc) 7. Use basic flip-flops 8. Analyze synchronous sequential circuits 9. Design synchronous sequential circuits 10. Operate registers and counters

	TENTATIVE COURSE OUTLINE					
	Week	Topics	Laboratory	Learning Outcome (LO)	Textbook Reading	Assignments / Exams
	1	Digital Systems and Binary Numbers		1	Chapter 1	
	2	Digital Systems and Binary Numbers, Boolean Algebra and Logic Gates		1,2,3	Chapter 1, Chapter 2	
	3	Boolean Algebra and Logic Gates		2,3	Chapter 2	
	4	Gate-Level Minimization		4,5	Chapter 3	
	5	Gate-Level Minimization	Experiment 1: Introduction to Logic Circuits	4,5	Chapter 3	Midterm1
	6	Gate-Level Minimization, Combinational Logic	Experiment 1: Introduction to Logic Circuits	4,5,6	Chapter 3, Chapter 4	
	7	Combinational Logic	Experiment 2: Logic Gates and Karnaugh Map	5,6	Chapter 4	
	8	Combinational Logic	Experiment 2: Logic Gates and Karnaugh Map	5,6	Chapter 4	
	9	Synchronous Sequential Logic	Experiment 3: Half Adder and its Applications	7,8,9	Chapter 5	
	10	Synchronous Sequential Logic	Experiment 3: Half Adder and its Applications	7,8,9	Chapter 5	Midterm2
	11	Synchronous Sequential Logic, Registers and Counters	Experiment 4: Decoders and Multiplexers	7,8,9,10	Chapter 5, Chapter 6	
	12	Registers and Counters	Experiment 4: Decoders and Multiplexers	9,10	Chapter 6	
	13	Registers and Counters	Experiment 5: Flip Flops and Sequential Circuits	9,10	Chapter 6	
	14	Memory and Programmable Logic	Experiment 5: Flip Flops and Sequential Circuits	9,10	Chapter 7	
	FINAL EXAMS WEEK (date and time to be announced later).					

COURSE ASSIGNMENTS	
A. Midterm Exams [40 %]	
There will be 2 midterm exams, 20 % for each exam.	
B. Final [30%]	
There will be a cumulative closed-book final exam covering all topics. Date and time of the final will be announced at the end of the semester.	
C. Lab performance [25%]	
There will be 5 experiments, 5% for each experiment.	
D. Project Assignment [5 %]	
Topics and details will be announced later.	

COURSE ASSESSMENTS & LEARNING OUTCOMES MATRIX	
Assessment Methods	Course Learning Outcomes
Lab Performance	LO # 1..10
Project Assignment	LO # 5,9
1 st Midterm Exam	LO # 1,2,3
2 nd Midterm Exam	LO # 4,5,6,7
Final Exam	LO # 1..10

Teaching Methods & Learning Activities	☒ Telling/Explaining ☒ Questioning ☒ Reading ☒ Demonstrating ☒ Problem Solving	☒ Collaborating ☒ Experiments ☒ Oral Presentations/Reports ☒ Web Searching ☒ Project Assignments
Assessment Methods (Formal & Informal)	☒ Test/Exam ☒ Quiz	☒ Presentation (Oral) ☒ Project Assignments
Student Workload (Total 160 Hrs)	☒ Lab Applications..... 36 hrs ☒ Course Readings 35 hrs ☒ Exams/Quizzes 36 hrs	☒ Project Report 10 hrs ☒ Project Assignment..... 15 hrs ☒ Lectures 28 hrs

COURSE POLICIES

I. Attendance

- Regular class attendance is expected for all students at the University. You are not required but advised to attend all classes.
- Please sign the attendance sheet when you come to the class. Any false signatures will result in zero participation grades for all parties involved.
- Your absence will not reduce your attendance rate *if and only if* you have a legitimate reason for missing a class (such as illness, death in family, a traffic accident, etc.). In case of an illness or emergency, you must supply a formal documentation that supports your claim.

II. Make-up Exams

Make-ups for Midterm Exams 1 and 2 will be available *if and only if* you have a legitimate reason for missing the exam (such as illness, death in family, a traffic accident, etc.). In case of an illness or emergency, you must supply a formal documentation that supports your claim.

III. Late Submission Policy

Late submissions will not be graded. There will be *no* make-up for quizzes and project assignments. Missed assignments and quizzes will result in a grade of zero (0).

IV. Cheating & Plagiarism

Collaboration is strongly encouraged; however, the work you hand in must be solely your own. Cheating and plagiarism are very serious offenses and will be penalized accordingly by the university disciplinary committee.

Cheating has a very broad description which can be summarized as "acting dishonestly". Some of the things that can be considered as cheating are the following:

- Copying answers on exams, projects and lab works,
- Using prohibited material on exams,
- Lying to gain any type of advantage in class,
- Providing false, modified or forged data in a report,
- Plagiarising (see below),
- Modifying graded material to be re-graded,
- Causing harm to colleagues by distributing false information about an exam, homework or lab.

All of the following are considered plagiarism:

- Turning in someone else's work as your own,
- Copying words or ideas from someone else without giving credit,
- Failing to put a quotation in quotation marks,
- Giving incorrect information about the source of a quotation,
- Changing words but copying the sentence structure of a source without giving credit,
- Copying so many words or ideas from a source that it makes up the majority of your work, whether you give credit or not.

(www.plagiarism.org)

V. Disability Support

If you have a disabling condition which may interfere with your ability to successfully complete this course, please contact Dr. Aslı Bugay (email: asli.bugay@tedu.edu.tr) or Dr. Tolga İnan (email: tolga.inan@tedu.edu.tr). For more information please see Handbook for Registered Students.

***** GOOD LUCK *****